# 1 Introduction to Analog Electronics and Circuits

## INTRODUCTION TO ANALOG CIRCUITS

After going through the course on electronic devices and circuits, we are now in a position to talk about analog electronics. But the question arises here is that, what is analog electronics? This is one of the questions which comes in the mind of learners before starting analog electronics. But an important question which should come would be Why analog electronics? Why should we study analog electronics?

In this chapter itself, we will try to find out the answer to the above questions.

**Rack Your Brain**

Why silicon is preferred over germanium?

## ANALOG ELECTRONICS (ANALOG + ELECTRONICS)

**1) Analog**

All the signals which are found naturally are basically analog signals, since, it is analogous to physical phenomena that they represent, are known as analog signals. Range of analog signals is from $-\infty$ to $+\infty$. Analog signals can take any value.

**Example:** Why speech signal is considered as an analog signal?

- Waveform of speech signal may look like as shown in the below figure:

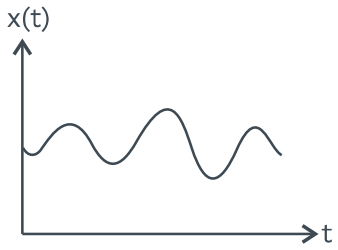



**Fig. 1.1 An Analog Signal**

- Amplitude at any time depends on the intensity of sound, which can take any value at any particular instant.

**Grey Matter Alert!**

Circuit has closed path and network may or may not have closed path.

**2) Electronics**

- Electronics is the study of flow of charge carriers (electrons and holes) in a gas, vacuum or semiconductor.
- Why it is important to study motion of electrons in semiconductor?
- It is important because we can control the flow of electrons in a semiconductor by appropriate doping.
- Therefore, if we control the flow of charge carriers, then we can control the flow of current and this phenomena makes revolution in field of semiconductors. Now a days, it has vast applications in our day-to-day life.

- **Discrete signals**
  Sometimes we don't require all the values of the given analog signal, but only require the values at a fixed instant of time and for this we use a process known as sampling.

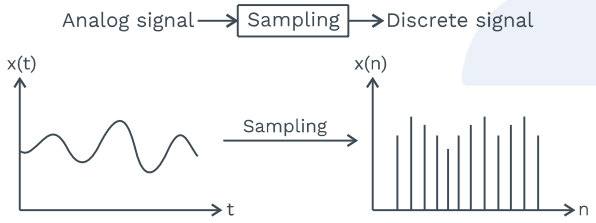



**Fig. 1.2 Analog signal** **Fig. 1.3 Discrete signal**

**Note:**
The number of samples depend upon the sampling rate at which analog signal is sampled.

- **Digital signals**
  Digital signals is obtained from discrete signals by quantizing discrete time signals followed by the encoding.

**Rack Your Brain**

What are the advantages of using discrete type signals?

**Grey Matter Alert!**

It is very difficult to change the conductivity of metals and insulators.

**Rack Your Brain**

Why electron mobility is higher than hole mobility?

## ANALOG CIRCUITS

The circuits which process analog signals as input and output are known as analog circuits. In other words, the circuits in which input as well as output signals are analog in nature are known as analog circuits.

- **Why study of analog circuits is important?**

As the real world is analog and all the physically occurring signals in any electronic circuit are always analog in nature. That's why the study of analog circuits is important. How these signals are usually processed in any circuit? We will study these concepts in this subject.

In this subject, we will also study about semiconductor devices such as diodes, transistors, operational amplifiers and their vast applications such as oscillators and amplifiers.

**Grey Matter Alert!**

All electronic circuits require DC power supply for their operation.

**Grey Matter Alert!**

Basis components of analog and digital circuits are same.

## CHAPTER SUMMARY



- Analog means analogous, which means signal which is analogous to naturally occurring physical phenomenon is known as analog signal. It is usually denoted by x(t).
- Discrete means fixed which means the signals which exist at fixed instant of time are known as discrete signals. They are usually denoted by x(n).
- Discrete signals are obtained from analog signals by a process known as sampling.
- Digital signals are the signals which have only two levels of value either high (1) or low (0).
- Electronics is the study of flow of charge carriers in gas, vacuum or semiconductor.
- The main advantage of semiconductor devices is that their electrical conductivity can be tuned.
- The circuits which process analog signals are known as analog circuits.
- Study of analog circuits is important because our real world is analog in nature and all the physically occurring signals are analog in nature.

## PRACTICE QUESTIONS

**Q1** **Which of the following is true regarding an analog signal?**
**a) It takes only discrete values.**
**b) It does not exist naturally.**
**c) It can take any value ranging from $-\infty$ to $+\infty$.**
**d) All the above are true.**

**Q2** **Which of the following is done to obtain discrete signal from an analog signal?**
**a) Sampling only**
**b) Quantizing only**
**c) Encoding only**
**d) Sampling and quantizing**

**Q3** **Which of the following is not a semiconductor device?**
**a) SCR** **b) Triode** **c) BJT** **d) None of above**

**Q4** **Which of the following is the most important feature in a semiconductor?**
**a) It has conductivity higher than insulator.**
**b) It has conductivity less than metal.**
**c) It has conductivity more than metal.**
**d) It's electrical conductivity can be changed.**

**Q5** **Which of the following cannot be done to change the conductivity of semiconductors?**
**a) Doping**
**b) Changing temperature**
**c) Add two suitable group elements**
**d) Using insulator doping**

**Q6** An analog signal is one in which
a) Amplitude is continuous
b) Time is continuous
c) Both amplitude and time must be continuous
d) Both amplitude and time must be discontinuous

**Q7** Human voice signal is usually
a) Digital
b) Analog
c) Discrete
d) Both analog and digital

**Q8** A digital signal is one in which
a) Amplitude is continuous
b) Time is continuous
c) Both amplitude and time must be continuous
d) Both amplitude and time must be discontinuous

**Answer Key:**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1) | c) | 2) | a) | 3) | d) | 4) | d) | 5) | d) |
| 6) | c) | 7) | b) | 8) | d) | | | | |